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PI Name:	Fernandez-Pello, Carlos Ph.D.		
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Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
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PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
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Abstract The objective of the program is to develop a new flammability apparatus and a bench-scale flammability test, FIST (Forced Ignition and Spread Test), that reproduces the environment that is expected in a manned space-based facility, i.e., microgravity and low velocity gas flows of variable oxygen concentration. The FIST would subsequently be used to obtain “flammability diagrams” of practical materials (plastics, composites, etc.) that are expected in these facilities. The research could eventually provide NASA with a test methodology to complement the existing protocol and provide a more comprehensive assessment of material flammability for space applications. Theoretical modeling, an extensive normal gravity data bank, and a few validation space experiments will support the testing methodology. This protocol could lead to better determination and ranking of the fire hazard characteristics of potential materials to be used in space-based environments, which are necessary criteria for an adequate fire protection design for space-based facilities. Task Description
The FIST apparatus is similar to the ASTM E-1321 LIFT apparatus, except the fuel sample is mounted in small-scale

Task Description:	wind tunnel so that a low-velocity oxidizer flow of specified oxygen concentration can be forced over the sample surface during the test. The oxidizer (air or a N2/O2 mixture of prescribed oxygen concentration) is forced through a rectangular duct at a known velocity. The duct is 56 cm in length, 10 cm high, and 12.7 cm wide. Its two side walls are clear Pyrex and its bottom surface is an insulating substrate (mica). A 30 mm by 30 mm test specimen is embedded flush in the substrate so that the boundary layer is not disrupted. Two radiant heaters are used to irradiate the sample with an approximately uniform heat flux of known intensity. Ignition is induced at fuel sample's trailing edge with an electrically heated Kanthal wire coil that acts as a pilot. Measured quantities for a particular material include the time to ignition as a function of ambient oxygen concentration and flow velocity, as well as the critical (minimum) heat flux for ignition. Task Objective The proposed research program will study the effects of microgravity on the flammability diagrams of solid combustible materials, and the fire properties derived from them. The final objective of the research is to develop a new flammability apparatus that would better reflect the potential ambient conditions of space-based environments, i.e., microgravity, low velocity flow, oxygen concentrations slightly below and above that of air, and use it to obtain more appropriate flammability diagrams for practical materials (plastics, cable jackets, electronic circuit boards, etc.) that will be found in space-based facilities. Task Significance The strong possibility of an accidental fire occurring in a space-based facility during a long mission, is a primary safety concern, and consequently it is of critical importance to characterize the fire properties of materials that are used.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Composites are finding increased use in the transportation and construction industries. One barrier to their widespread adoption is the increased flammability of polymer based composites when compared with other materials, particularly metals and concrete. In the process of developing a test methodolgy for flammability of composites intended for use in space structures, the FIST project is developing a better understanding of composite flammability useful to ground based design and manufacture.
	Experimental Subtask Ignition delay: A type K thermocouple welded to the igniter has allowed for direct thermocouple measurements of igniter temperatures. These measurements have proven to be more stable and reproducible than the IR camera measurements conducted previously. A series of tests using these measurements have provided a minimum igniter temperature of 1105 C above which no further changes in ignition delay time is seen with increases in igniter temperature. A series of tests were run to examine the effect of oxygen concentration on the ignition delay time of PP/30%GL. These tests were conducted at a flow rate of 1 m/s, at O2 concentrations of 17-24% and under variable heat fluxes. Tests showed an increase in ignition delay times with decreases in oxygen concentration and a limiting oxygen index for lower heat fluxes. Test results will be used to draw the ignition delay branch of flammability diagrams under variable oxygen concentrations and determine the limiting oxygen index for PPG for a range of heat fluxes. Weight loss measurements of PP/30%GL and PMMA flame spread samples and ignition delay samples were forwarded to NASA GRC for use in filter design. This work led to development of a continuous mass loss measurement system, which measures mass changes in samples during the heating, ignition and mass burning phases of ignition delay tests. Work to date has focused on development of a Phlogopite Mica sample holder, which accommodates the polypropylene and polymethylmethacrylate samples while providing stability during mass loss measurements and thermal insulating conditions similar to those in current use in the FIST development tunnel at NASA GRC. These tests are used to obtain information on the critical mass flux for ignition, examine the protective effects of blowing on oxidative pyrolysis and heating of the surface, support the numerical modeling of the FIST process through determination of mass loss rate, surface temperature, and back face temperature histories and to provide more accurate information about total mass loss for use in filter design. Analysis of the mass flux rate at the time of ignition gives good repeatability and has demonstrated an increase in mass flux rate at ignition with increasing heat flux for PPG. There does not appear to be a strong relationship between oxygen concentration in the induced flow and critical mass flux at ignition and work is underway to examine the effect of oxidizer velocity on critical mass flux at ignition. A series of KC-135 tests of ignition delay with samples in the intended configuration for flight were completed. Examination of thermocouple readings showed consistency with microgravity tests completed on the KC-135 with larger sample sizes. IR and video imagery provided information on necessary improvements to the image acquisition systems intended for flight. The UC Berkeley lab purchased a supply of black type G PMMA and PP/30%GL (PF062-3). A portion of each material has been forwarded to NASA GRC to maintain material consistency between NASA GRC and UC Berkeley. The FIST tunnel has been modified with a particle seeding system and mounting hardware to accommodate a particle image velocimetry (PIV) system. Flame spread: The FIST tunnel has been modified allowing for easier transitions to the flame-spread configuration. PP/30%GL and PMMA flame spread tests were filmed using an IR camera and these digital IR images were processed using the SPOTLIGHT 1.0 image analysis software provided by NASA. This ground based analysis demonstrated that the SPOTLIGHT software allows for velocity measurements of the flame front under a variable heat flux while quantifying surface temperatures of the fuel ahead of the flame front. Flame spread tests conducted on the KC-135 were evaluated using the methods developed in ground tests using the SPOTLIGHT software. Flame spread velocities for periods of microgravity and the 1.8 g pull-up period were quantified and analysis of IR imagery has provided information on necessary improvements to the image acquisition system. Theoretical Subtask As reported in FY03, a model that calculates the transient heating and gasification of a solid material exposed to thermal radiation has been combined with a model that provides a detailed description of phenomena occurring in the gas phase. A Computational Fluid Dynamics (CFD) code developed by the National Institute of Standards and Technology (NIST) known as Fire Dynamics Simulator (FDS) was used as the starting point. The FDS code numerically solves a discretized

Task Progress:	form of the “mildly compressible” Navier Stokes equations, which have come to be called the “low Mach number combustion equations”. In this way, a detailed spatial description of the flowfield (velocity, temperature, heat release) is obtained. The combined solid and gas phase model was used to predict the ignition time of polypropylene/30% glass composite samples in normal and reduced gravity, and the results were compared with available experimental data. The reduced-gravity data were obtained during an earlier KC-135 flight campaign. Good agreement between the prediction and experiment was obtained. Both the experiment and prediction suggest that materials are easier to ignite in reduced-gravity low speed flows such as those that exist in manned space-based facilities. The numerical predictions also indicate that the pyrolysis rate and surface temperature at ignition are both considerably lower in low speed microgravity flows than in normal gravity flows where the minimum velocity is limited by buoyancy. This has important fire safety implications because it suggests that solid combustibles are easier to ignite under the conditions expected in space-based facilities (low-velocity ambient oxidizer flows and reduced gravity). These findings were submitted to the journal Combustion Science Technology. The paper was accepted and is currently in press. The model was also used to examine the influence of ambient oxygen concentration on the calculated ignition delay time for the polypropylene 30% glass composite material. As a first approximation, the effect of oxygen concentration on the solid-phase decomposition rate was not included. Thus, any change in the predicted ignition time with ambient oxygen concentration is solely due to gas-phase effects. It was found that the model’s predictions are quite insensitive to ambient oxygen concentrations over the range 18%-30%, whereas the experimental measurements show that the ignition time decreases as ambient oxygen increases. However, recent mass loss rate measurements in our laboratory (as well as several papers in the literature) suggest that the rate of solid-phase polypropylene degradation is quite sensitive to the ambient oxygen concentration, increasing with oxygen level. As mentioned above, this phenomenon (more rapid decomposition at higher oxygen levels) has not yet been included in the model but could be approximated quite simply using a solid-phase activation energy that decreases with ambient O2 level. Recent experimental mass loss rate measurements made in our laboratory suggest that the freestream oxygen concentration influences the ignition time mostly by increasing the rate at which the solid material decomposes rather than through gas-phase effects because the mass loss rate at ignition was found to be relatively insensitive to ambient oxygen concentration. This is consistent with the modeling results. The combined solid/phase and gas-phase model was used to calculate opposed flow flame spread rates on polypropylene/glass and polymethylmethacrylate samples. The available experimental data and trends are captured qualitatively by the model. A comprehensive comparison of model predictions to experimental data has not yet been performed, but is planned for FY05. The experimental data sets will include normal-gravity ground-based measurements made in our laboratory as well as microgravity flame spread data obtained on KC-135 flights in October 2004. During FY04, several important improvements were made to the solid-phase model. It was modified to use a highly efficient recursive algorithm for solving the heat conduction equation. The computational cost of the model was reduced by several orders of magnitude, particularly for thick materials where several hundred cells may be required. The new algorithm was verified by comparison with an analytical solution for a semi-infinite solid exposed to a constant net heat flux. The reduced computational cost of the new model makes it possible to apply stochastic search methods for optimization of input parameters. With this approach, the thermal properties and decomposition kinetics of a particular material can be estimated by finding the set of properties that minimizes the error between experimental measurements and the model predictions. However, several hundred thousand trial solutions are typically required, and this approach was not practical with earlier versions of the solid-phase model due to computational cost. An optimization technique was developed that uses a self-adaptive genetic algorithm based on the principles of Darwinian natural selection in combination with the solid phase-pyrolysis model. The genetic algorithm is essentially an optimization tool that is used to find the combination of material properties that provides the best agreement between model prediction and experiment. The required experimental measurements are mass loss rate, surface temperature, and back face temperature histories. Preliminary results are very encouraging. This method will allow us to apply the solid-phase model to other solids for which the required material properties are unknown and not available in the literature because they can be inferred from experiments in the FIST apparatus. Finally, work was started to extend the existing solid-phase model to handle charring and intumescent materials that may actually be used in extraterrestrial habitats. Due to the increased complexity of the underlying physical models for charring materials, the number of required material properties will be increased. However, the genetic algorithm optimization technique should make it possible to estimate the required material properties, even for complex charring or intumescent materials.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
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